STAT 107 Final Report

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Analyzing the Post-Pandemic Fluctuation of Metro Rail Ridership in Los Angeles

The COVID-19 pandemic significantly disrupted public transportation usage across Los Angeles County, causing a substantial decline in LA Metro rail ridership. This project examines how rail ridership has changed since 2020, comparing post-pandemic trends with pre-pandemic levels to evaluate the extent of recovery. In particular, we investigate whether recent infrastructure developments have corresponded with noticeable changes in ridership. Our analysis aims to identify patterns, fluctuations, and potential relationships between transportation infrastructure improvements and public transit demand. Preliminary observations suggest that although ridership has gradually recovered since the pandemic, it continues to face challenges in returning to pre-pandemic levels. By examining these trends, this project seeks to better understand the factors influencing public transportation recovery and identify opportunities to encourage greater rail transit usage throughout Los Angeles County.

Introduction

The central question this project aims to answer is, "How has LA Metro rail ridership in Los Angeles County changed since the COVID-19 pandemic, and to what extent have recent developments influenced its recovery compared to pre-pandemic levels?"

To investigate this question, we will analyze LA Metro rail ridership data, examining changes from pre-pandemic til' present time. We will compare ridership levels, identify upward trends, and investigate fluctuations that may correspond with major infrastructure developments, including but not limited to, the D Line extension and the opening of the LAX/Metro Transit Center.

This analysis may benefit LA Metro transportation planners, local government officials, policymakers, and Los Angeles County residents. Understanding how ridership responds to infrastructure improvements and an overall increase in area covered, would greatly help inform future public transportation investments and transit policies.

Rather than developing a predictive model, our primary objective is to conduct an exploratory data analysis and further investigate potential relationships between infrastructure developments and ridership recovery. Ultimately, we hope to identify trends that can help explain trajectroy of post-pandemic public transportation usage and highlight potential opportunities to increase rail ridership in the future.

Here is the link to the repository of our project on GitHub:

https://github.com/adhafawwaz/STAT107_FinalProject_AdhaDheekshawMelvin.git

Executive Summary

1. Significant pandemic decline: LA Metro rail ridership has experienced a substantial and steep decrease beginning in March 2020, reflecting changes in commuting habits and reduced public transporation

demand during the COVID-19 pandemic.
2. Gradual post-pandemic recovery: Ridership has generally demonstrated an upward trend following the initial pandemic decline. However, recovery has been gradual and subject to fluctuations.
3. Infrastructure developments: Recent transportation projects, such as LAX/Metro Transit Center, Regional Connector, and the D Line extension represent relevant developments that may influence rail accessibility and ridership patterns.
4. Implications for Future Transit Planning: Understanding the relationship between infrastructure improvements, commuting patterns, and ridership demand may help policymakers identify strategies to encourage greater public transportation usage throughout Los Angeles County.

Data

For this project, we are using LA Metro rail ridership data from 2019 to 2026. We downloaded this data as a CSV file from ridership.streetsforall.org. It contains 534 observations and six variables: line_name, year, month, est_wkday_ridership, est_sat_ridership, est_sun_ridership.

The dataset includes the rail line, year, month, and estimated ridership for weekdays, Saturdays, and Sundays. We used this information in order to study how the rail ridership has changed over time, specifically before, during and after the COVID-19 pandemic.

We imported the dataset into R, ensuring we had data from 2019 and after. We checked the variables and data types, and also checked for missing values and duplicate observations. We also checked for zero and negative ridership values. We kept the zero values because they might represent the times when certain lines were not in operation, but we are yet to confirm this.

We also made two functions in R to help with cleaning the data. The first, add_date(), combines the year and month into a date column. The second, check_zeros(), checks for zero values in the ridership data. This helps us check the data and prepare it for our graphs.

Finally, we created a date column by combining the year and month. This makes it easier to create graphs showing how ridership changed over time. We did not remove any observations during this initial cleaning process, because our dataset was comprehensive and did not include easily-identifiable inconsistencies.

Visualization

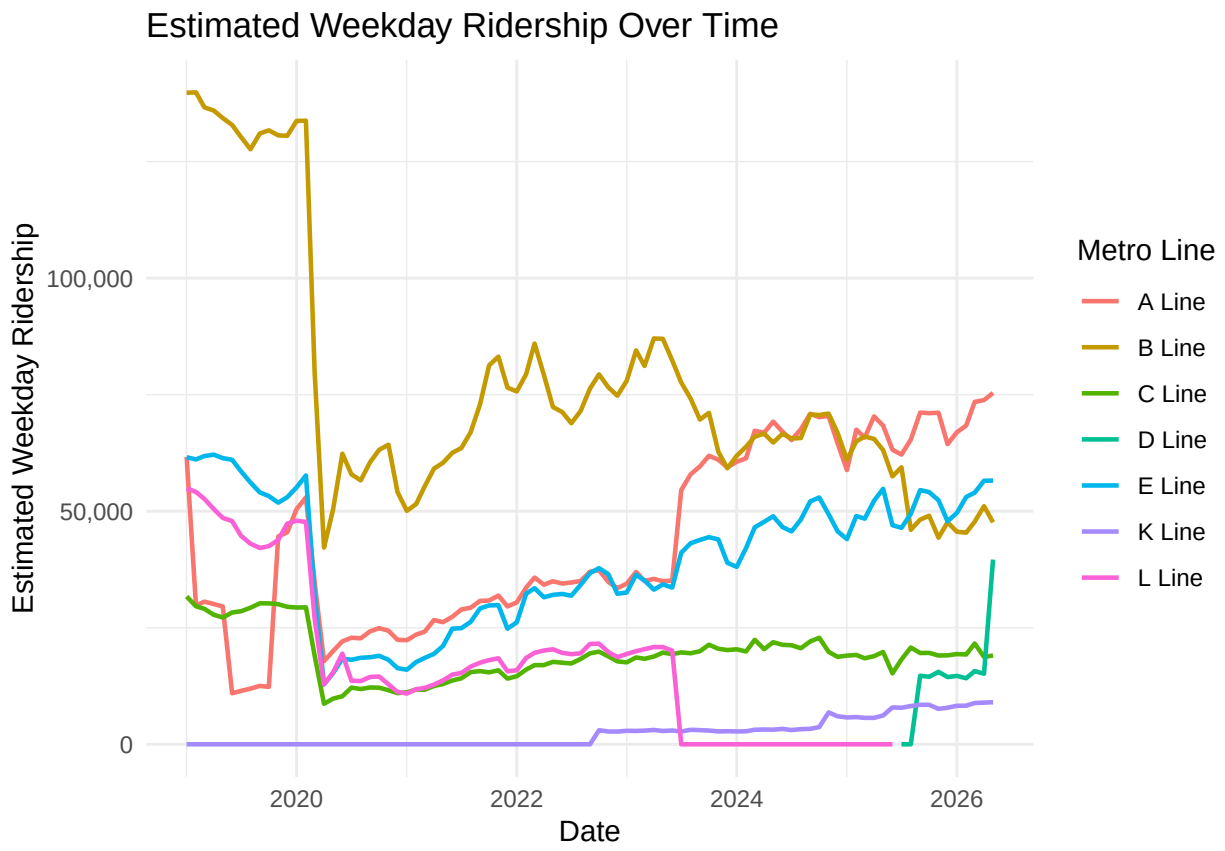
```
## Warning: package 'dplyr' was built under R version 4.5.2
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##
## Attaching package: 'dplyr'
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## The following objects are masked from 'package:stats':
##
##     filter, lag
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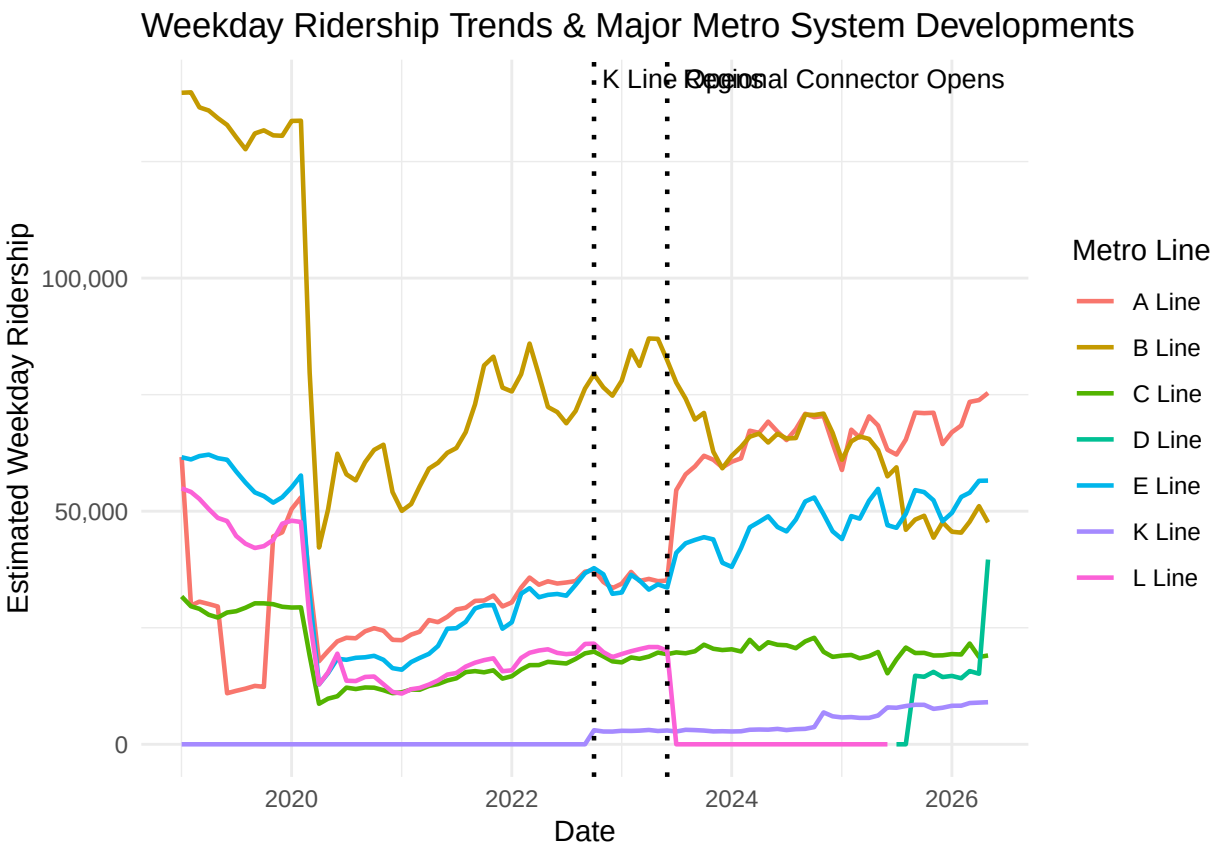
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## The following objects are masked from 'package:base':
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##     intersect, setdiff, setequal, union
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## Warning: package 'ggplot2' was built under R version 4.5.2
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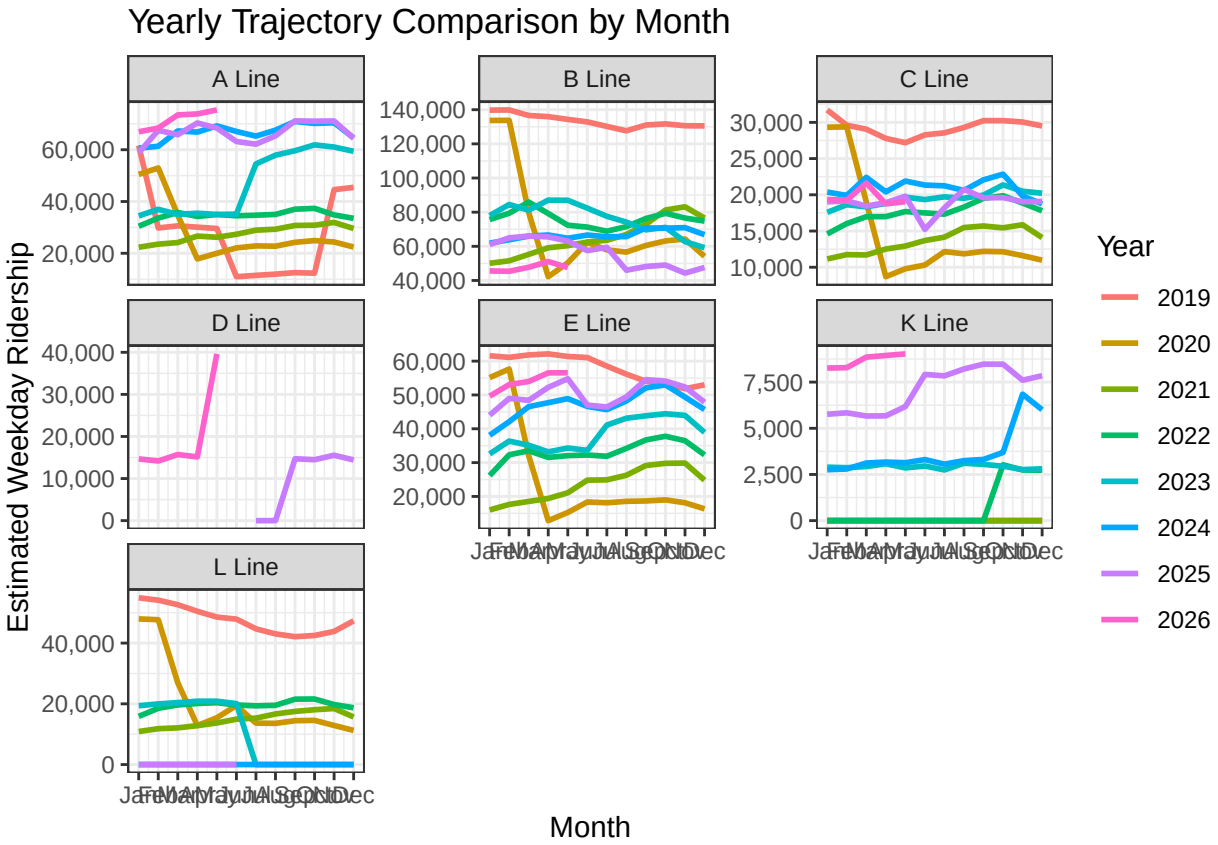
1. First Line Graph

This line graph shows monthly weekday ridership for each LA Metro rail line from 2019 to 2026. Before 2020, the B Line had the most riders, with around 100,000 daily boardings, while the A and E Lines had about 50,000 to 60,000. In March 2020, ridership dropped by more than 70% across all lines because of the pandemic. Ridership slowly started to recover after 2021, but things changed around mid-2023. The B Line's recovery slowed down, while the A and E Lines saw a big increase in riders. By 2024–2026, the A Line had caught up to and even passed the B Line in weekday ridership.



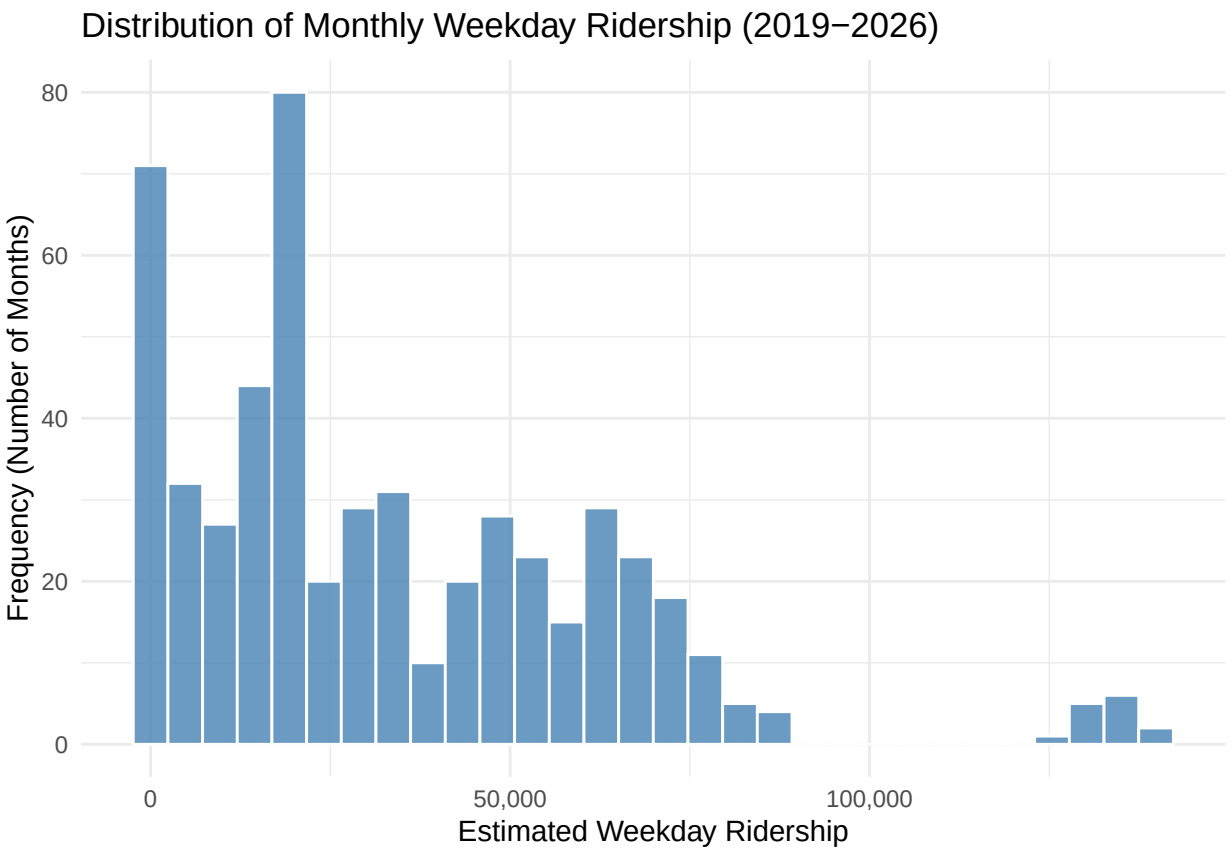
2. Second Line Graph

The second line graph shows the same monthly weekday ridership trends from 2019 to 2026, but it also includes vertical lines to mark when new rail lines or extensions opened. These lines help show how new projects may have affected ridership over time. In particular, the June 2023 Regional Connector opening lines up with the increase in A and E Line ridership. This suggests that the new connection may have helped bring in more riders and changed how people travel across the Metro system.



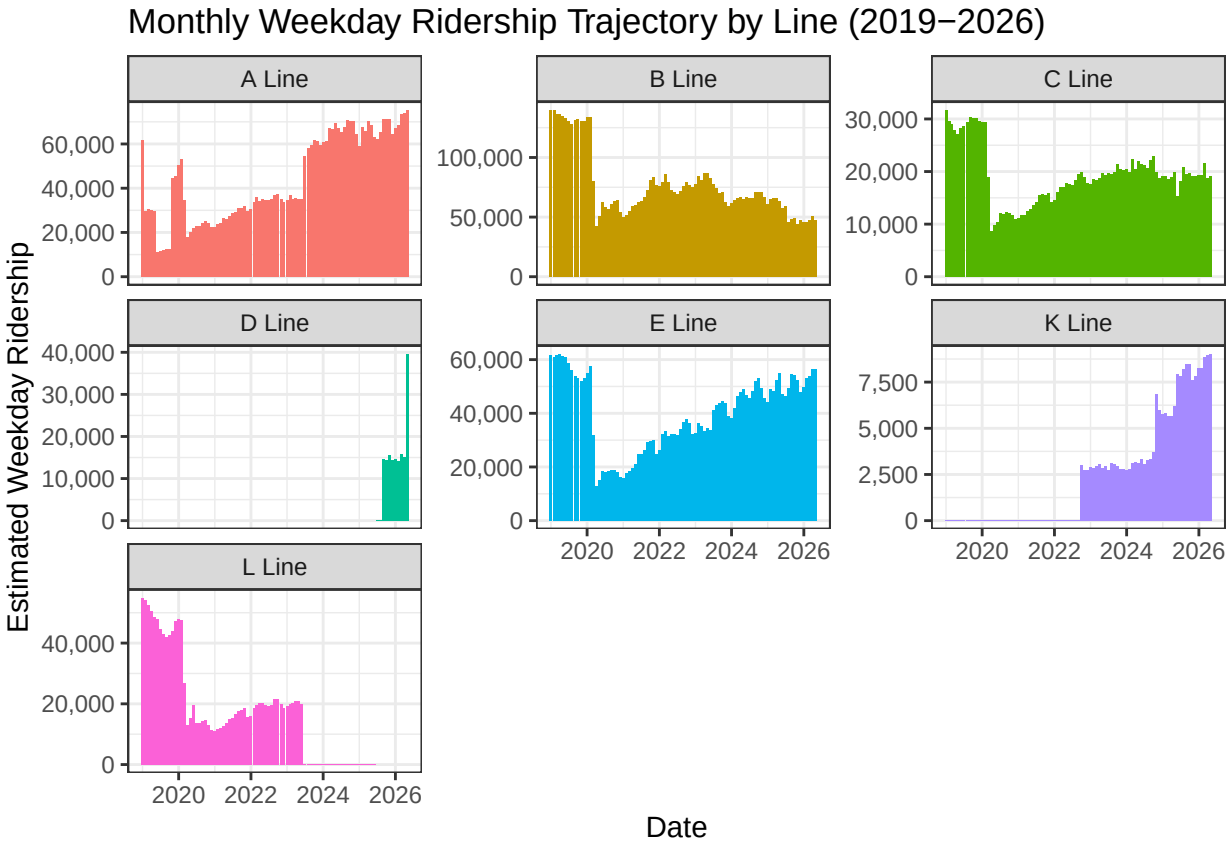
3. Faceted Line Graph

This line graph compares monthly weekday ridership for each Metro line from 2019 to 2026. All lines show a big drop in 2020 because of the pandemic, but their recovery looks different. The B and D Lines have stayed below their 2019 ridership levels, likely because more people continue to work from home. On the other hand, the A and E Lines started seeing strong increases around mid-2023 and eventually reached or passed their pre-pandemic levels. This could be connected to the June 2023 Regional Connector opening, which helped improve light rail connections across LA County.



4. Histogram

This histogram shows how monthly weekday ridership is spread across all LA Metro rail lines from 2019 to 2026. The graph has two main peaks. The first peak represents lower ridership numbers after the pandemic, along with lines that normally have fewer riders. The second peak represents higher ridership levels from before the pandemic and the recent recovery of some light rail lines. This shows that ridership has changed over time, meaning we need to consider both the pandemic and new rail expansions when studying how the system has recovered.



5. Faceted Bar Chart

This bar chart shows monthly weekday ridership for each Metro line from 2019 to 2026. All lines had a major drop in early 2020 because of the pandemic, but they have recovered at different rates. The B and D Lines are still below their pre-pandemic ridership levels, possibly because more people are working from home. Meanwhile, the A and E Lines have seen steady growth since mid-2023, with ridership eventually reaching or passing 2019 levels. This growth may be linked to the June 2023 Regional Connector opening, which made it easier for people to trave

Analysis

First, we will clean and organize the dataset by checking for inconsistencies, duplicate observations, etc. We will then use descriptive statistics in order to summarize ridership patterns across time periods. In particular, we will compare pre-pandemic ridership (using 2019 as our baseline) with ridership during and after the pandemic to measure the extent of recovery.

Next, we will plan to create data visualizations, such as line graphs, histograms, and bar graphs, to illustrate the ridership fluctuations across a period of time. These visualizations will help identify declines, recovery periods, and potential changes corresponding with infrastructure developments.

In addition, we may conduct hypothesis tests to determine whether the differences between pre-pandemic and post-pandemic ridership levels are statistically significant. We also plan to calculate percentage changes and recovery rates relative to 2019 to quantify how closely ridership has returned to its past peaks.

Lastly, we hope to identify whether infrastructure changes correspond with noticeable increases in ridership. We expect to find that despite the fact ridership hasn't fully recovered past pre-pandemic levels, we would still anticipate improvements in transit accessiblity contributing to increased usage.